Two Entanglement Entropies of the Conjugate Weil Pair: the Admissible Pair Sector and the Full Residual Rank

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Abstract

The companion entanglement note and the orthogonality paper E1 identify two distinct entanglement entropies of the conjugate Weil pair $\{c, q-c\}$, which must not be conflated. The first is the entropy of the minimal admissible pair sector: the rank-3 admissible projection Π_q is spectrally atomic, equal to $\mathrm{span}\{e_0, e_{\xi_c}, e_{q-\xi_c}\}$; removing the zero mode leaves the two-dimensional Born–Infeld parity pair $\{e_{\xi_c}, e_{q-\xi_c}\}$, identified with the spin-$\frac{1}{2}$ representation. The proto-state on this sector is the singlet, and its reduced state is maximally mixed, so $S_{\mathrm{ent}}^{\mathrm{adm}} = \log 2$ for general canonical blocks, unconditionally. This is the sector relevant to the singlet correlator and the Tsirelson bound. The second is the full residual fibre-level rank $r_{\mathrm{pair}}(n) = R_\infty - R(n)$ defined on the full Gram–Schmidt basis (rank q at saturation, not the rank-3 admissible projection). E1 proves $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$ unconditionally for matched single-character blocks; for independently sampled canonical blocks it remains conditional on the residual admissibility indiscernibility hypothesis [$\mathrm{H_{res}}$]. The metaplectic dilation ϕ_ω is excluded as a bridge for [$\mathrm{H_{res}}$] because it acts between blocks, not within the residual support of one block.

Keywords: conjugate Weil pairs; spectral atomicity; admissible projection; spin-$\frac{1}{2}$ singlet; Tsirelson bound; residual rank; entanglement entropy; epistemic stratification.

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1 Introduction

The companion entanglement note [3] and the orthogonality paper E1 [5] establish results about the entanglement of a closed conjugate Weil pair $\{c, q-c\}$ in the spectral admissibility cascade. A careful reading shows that two genuinely different entanglement entropies are in play, attached to two different projections. Conflating them produces an apparent tension between an unconditional $\log 2$ and a depth-dependent $\log r_{\mathrm{pair}}(n)$.

This note separates them. The separation is not cosmetic: the two objects live on projections of different rank, have different physical roles, and have different epistemic status. One is the entropy of the minimal admissible sector, which is two-dimensional and carries the singlet relevant to Bell–Tsirelson correlations; it is unconditional for general blocks. The other is the entropy of the full residual Gram–Schmidt rank, a finer pipeline-level observable; it is unconditional only for matched blocks and otherwise rests on a bridge hypothesis.

2 Two distinct objects

The O-series pipeline stores two distinct objects per conjugate pair, and *O21* [7] (Remark 2.1) is explicit that they must not be identified.

- The full Gram–Schmidt basis is a unitary $q \times q$ matrix whose rows span all of $\mathbb{C}^q$ at saturation. Its cumulative admitted rank is $R(n)$, growing to $R_\infty = q$ at saturation, and the residual fibre-level rank is

  $$r_{\mathrm{pair}}(n) = R_\infty - R(n),$$

  which is non-increasing in n. This is the object of the companion note [3]. It is *not* the theoretical admissible projection.

- The theoretical admissible projection Π_q is the rank-3 projection onto the first three Gram–Schmidt directions $\{\phi_0, \phi_1, \phi_2\}$, identified through the H_{eff}-projection coordinates of *O21* [7], Remark 2.1. All spectral statements about Π_q concern this rank-3 object.

We therefore distinguish the admissible pair entropy $S_{\mathrm{ent}}^{\mathrm{adm}}$, defined on Π_q (Section 3), from the full residual-rank entropy $S_{\mathrm{ent}}(n)$, defined on the full Gram–Schmidt basis (Section 4).

3 The admissible pair sector: $S_{\text{ent}}^{\text{adm}} = \log 2$

The admissible sector is not an arbitrary residual subspace. Its structure is fixed by spectral atomicity.

Theorem 1 (Spectral atomicity, *O21* [7], Proposition 4.1)**.** *For each prime q and each conjugate pair $(c, q-c)$, the three admissible directions are pure Fourier modes,*

$$\phi_0 = e_0, \qquad \phi_1 = e_{\xi_c}, \qquad \phi_2 = e_{q-\xi_c},$$

with $e_\xi(k) = q^{-1/2} e^{2\pi i \xi k/q}$ and $\xi_c \in \{1, \ldots, \lfloor q/2 \rfloor\}$. Equivalently $\operatorname{ran}(\Pi_q^{(c)}) = \operatorname{span}\{e_0, e_{\xi_c}, e_{q-\xi_c}\}$.

This is an unconditional theorem of *O21*: it replaces the earlier coercivity and tightness hypotheses with proved statements, and is confirmed numerically to machine precision for $q \in \{29, 61, 101, 151\}$. Its proof rests on three closed steps: the k=3 Weil fingerprints are pure Fourier modes (the orthogonality result of E1 [5]); Gram–Schmidt preserves purity; and the admissible rank is exactly three [4], which with the Born–Infeld parity involution forcing e_{ξ_c} and $e_{q-\xi_c}$ to enter together fixes the triple.

The zero mode is removed by admissibility.

Lemma 2 (Mean-zero, *O21* [7], Lemma 3.1)**.** *For every $f \in \operatorname{ran}(\Pi_q^{(c)})$, the Fourier coefficient $\hat{f}(0) = 0$.*

Proof. The Weil representation ρ_c with $c \neq 0$ is irreducible with non-trivial central character, so the constant function (the trivial representation) is orthogonal to $\operatorname{ran}(\Pi_q^{(c)})$, giving $\langle \mathbf{1}, f \rangle = \hat{f}(0) = 0$. □

After removing e_0, the admissible sector is the two-dimensional Born–Infeld parity pair

$$\mathcal{A}_c := \operatorname{span}\{e_{\xi_c}, e_{q-\xi_c}\}, \qquad q - \xi_c \equiv -\xi_c \pmod q,$$

which we identify with the spin-$\frac{1}{2}$ representation $V_{1/2} \cong \mathbb{C}^2$. The two factors of the conjugate pair $\{c, q-c\}$ each carry a copy of $\mathcal{A}_c$, and the bipartite proto-state lives on $\mathcal{A}_c \otimes \mathcal{A}_c$.

Theorem 3 (Admissible pair entropy)**.** *The proto-state on $\mathcal{A}_c \otimes \mathcal{A}_c$ is the singlet*

$$|\Omega_c\rangle = \frac{1}{\sqrt{2}}\big(|\xi_c\rangle \otimes |q-\xi_c\rangle - |q-\xi_c\rangle \otimes |\xi_c\rangle\big).$$

Its reduced state is $\rho_c^{\mathrm{adm}} = \frac{1}{2}\mathbf{1}_2$*, and hence*

$$S_{\mathrm{ent}}^{\mathrm{adm}} = \log 2.$$

Proof. By Theorem 1 and Lemma 2, the admissible sector of each factor is $\mathcal{A}_c \cong V_{1/2}$. By *Q3* [8], Theorem 4.1, Born–Infeld indiscernibility forces the proto-state to be invariant under the diagonal action on $V_{1/2} \otimes V_{1/2}$, and by Schur's lemma applied to the Clebsch–Gordan decomposition $V_{1/2} \otimes V_{1/2} = V_0 \oplus V_1$, the unique normalised invariant vector is the singlet $|\Omega_c\rangle$. The Schmidt decomposition of $|\Omega_c\rangle$ has two terms with equal coefficients $1/\sqrt{2}$, so $\rho_c^{\mathrm{adm}} = \frac{1}{2}\mathbf{1}_2$ and $S_{\mathrm{ent}}^{\mathrm{adm}} = -\mathrm{Tr}(\frac{1}{2}\mathbf{1}_2 \log \frac{1}{2}\mathbf{1}_2) = \log 2$. □

Theorem 3 uses only the admissible parity pair $\{e_{\xi_c}, e_{q-\xi_c}\}$, fixed by spectral atomicity and parity for any block parameters. It does not require a canonical identification of the full residual supports. The admissible pair entropy is therefore $\log 2$ for general canonical blocks, unconditionally.

4 The full residual rank: matched blocks and $[\mathrm{H}_{\mathrm{res}}]$

The full residual fibre-level rank $r_{\mathrm{pair}}(n) = R_\infty - R(n)$ is a different object, defined on the full Gram–Schmidt basis rather than on Π_q.

On matched single-character blocks, E1 [5] (Theorem 7) proves that the residual Schmidt spectrum is flat and that

$$S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n).$$

The proof uses the sign-reflected Fourier supports $S_{q-c}^{\mathrm{res}}(n) = -S_c^{\mathrm{res}}(n)$, which canonically identify the residual modes of the two factors and yield a maximally entangled residual state.

For independently sampled canonical blocks, E1 also shows that this literal mode-by-mode identification is not automatic. *O17* [1] supplies equality of the residual dimensions of the two factors, but not a canonical pairing. Universal admission orthogonality ($w_j = 1$, E1 Theorem 4) ensures each factor has an orthonormal residual basis, but orthonormality and equal dimension do not by themselves fix the joint state or force a flat marginal.

Definition 1 (Residual admissibility indiscernibility $[\mathrm{H}_{\mathrm{res}}]$)**.** *At BFS depth n, the residual fingerprint directions of a conjugate Weil factor are residually admissibility-indiscernible if no invariant carried by the admissible generators distinguishes them, equivalently if assigning unequal weights would constitute a premature admissible resolution of the open residual fibre.*

Proposition 4 (Conditional flatness)**.** *Assume the closure of the conjugate pair as a pure object, and assume* $[\mathrm{H_{res}}]$*. Then the reduced state on the residual support is* $\rho_c^{\mathrm{res}}(n) = r_{\mathrm{pair}}(n)^{-1}\mathbf{1}_{r_{\mathrm{pair}}(n)}$*, and* $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$ *for general canonical blocks.*

Proof. Axiom A3 [2] and the non-factorisability lemma of the Foundation paper forbid a decomposition of the open fibre stable under the admissible generators prior to projection locking. A non-uniform diagonal reduced state separates the residual directions into distinct weight sectors. Under $[\mathrm{H_{res}}]$, the weight levels are not admissibility-visible invariants, so this separation is not a generator-stable decomposition and is admissible only if uniform. Together with closure (pure joint state) and the equal residual dimensions of *O17* [1], the reduced state is maximally mixed on the residual support, giving $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$. □

Remark 1 (The dilation does not supply $[\mathrm{H_{res}}]$)**.** *The metaplectic dilation* $\phi_\omega(a, b, z) = (\omega a, \omega^{-1}b, z)$ *of* O31 *[6] forces* $\sigma_c(n) = \sigma_{\omega c}(n)$*, relating the block of character* c *to the block of character* ωc*. It acts between blocks, not within the residual support of a single block, and so does not permute the residual modes of one factor. It cannot make the residual modes admissibility-indiscernible, and is therefore not a bridge for* $[\mathrm{H_{res}}]$*.* $[\mathrm{H_{res}}]$ *has a status parallel to the conditional hypotheses [H-color] and [H-ext] elsewhere in the programme.*

5 Relation to the Bell–Tsirelson sector

The entanglement that Bell–CHSH experiments measure is carried by the spin-$\frac{1}{2}$ singlet. By Theorem 3, this is exactly the admissible pair sector, with $S_{\mathrm{ent}}^{\mathrm{adm}} = \log 2$. The singlet correlator $E(\hat{a}, \hat{b}) = -\hat{a} \cdot \hat{b}$ and the unconditional Tsirelson bound $|S_{\mathrm{CHSH}}| \leq 2\sqrt{2}$ are derived from the same admissible structure in *Q1* [2] (Theorem 2.14 and Corollary 2.16). The admissible pair entropy is therefore the entanglement entropy of the state behind those results.

The full residual-rank entropy $S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$ is a finer, pipeline-level observable that tracks the contraction of the full Gram–Schmidt span toward saturation. It is not the quantity measured by a Bell experiment, and it should not be identified with the singlet entropy. The two coincide only in the degenerate case $r_{\mathrm{pair}}(n) = 2$.

6 Status

E1 [5] proves universal admission orthogonality: each admitted k=3 Weil fingerprint is a Fourier mode, and a newly admitted frequency is orthogonal to the previous span. *O17* [1] proves equality of the conjugate pair ranks.

The admissible spin-$\frac{1}{2}$ pair sector is closed by spectral atomicity (*O21* [7], Proposition 4.1), the mean-zero lemma, and the singlet identification (*Q3* [8], Theorem 4.1). The zero mode is excluded and the two surviving non-zero frequencies form the parity pair $\{e_{\xi_c}, e_{q-\xi_c}\}$. Therefore

$$S_{\mathrm{ent}}^{\mathrm{adm}} = \log 2$$

for general canonical blocks, unconditionally (the rank-3 input of *O29* [4] carries the analytical backing of *O30* [6]).

By contrast, the full residual-rank statement

$$S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n)$$

is unconditional only for matched single-character blocks (E1 Theorem 7). For independently sampled canonical blocks it remains conditional on $[\mathrm{H}_{\mathrm{res}}]$, the residual admissibility indiscernibility hypothesis (Definition 1). The metaplectic dilation is excluded as a bridge for $[\mathrm{H}_{\mathrm{res}}]$ (Remark 1).

The separation is therefore:

$$\begin{aligned} \text{Bell–Tsirelson sector} &\implies S_{\mathrm{ent}}^{\mathrm{adm}} = \log 2, \\ \text{full residual-rank sector} &\implies S_{\mathrm{ent}}(n) = \log r_{\mathrm{pair}}(n). \end{aligned}$$

The first holds unconditionally for general canonical blocks. The second holds unconditionally only for matched single-character blocks, and otherwise rests on $[\mathrm{H}_{\mathrm{res}}]$.

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